Using OpenCV *for* Machine Learning in Real-time Computer Vision *and* Image Processing

Catch up with Open Source Computer Vision (OpenCV), a computer vision and machine learning software library. As it is published under the BSD licence, you are free to develop and modify the source code.

Computer vision and digital image processing are currently being widely applied in face recognition, biometric validations, the Internet of Things (IoT), criminal investigation, signature pattern detection in banking, digital documents analysis, smart tag based vehicles for recognition at toll plazas, etc. All these applications use image and real-time video processing so that the live capture of multimedia impressions can be made for detailed analysis and predictions.

Computer vision is widely integrated in different applications including 2D and 3D image analytics, egomotion estimation, feature points detection, human-computer interaction (HCI), face recognition systems and mobile robotics. Computer vision is also applied in the fields of gesture recognition, object identification, segmentation recognition, motion understanding, stereopsis stereo vision, motion tracking, structure from motion (SFM), pattern recognition, augmented reality, decision making, scene reconstruction, etc.

The following are the key research areas in computer vision and image analytics:

- Deep neural networks for biometric evaluation
- Facial sentiment analysis and emotion recognition
- Real-time video analysis and recognition of key features
- Gesture recognition
- Visual representation learning
- Face smile detection
- Vein investigation and training for biometrics
- Multi-resolution approaches
- Radiomics analytics for medical data sets
- Forgery detection in digital images
- Content based image retrieval
- Automatic enhancement of images
- Identification and classification of objects in real-time
- Defects prediction in manufacturing lines using live images of machines
- Industrial robots with real-time vision for disaster management
- Image reconstruction and restoration
- Computational photography
- Morphological image processing
- Animate vision
- Photogrammetry

Prominent tools for computer vision and image analytics

There are a number of tools and libraries available to implement